

Reproduced results from the two thermoelectric dataset papers

Summary

I have reproduced both papers' dataset curation methods and checked the numbers against what's published, using the authors' own code and data wherever I could get hold of it. For Paper 1 (Ghosh & Tewari, IIT Roorkee), that's the full dataset analysis section, Figs. 6 to 13. For Paper 2 (Athar, Mecibah & Jund, Montpellier/ICGM), that's the round-robin filtering method itself, run on their actual GitLab script.

Paper 1: Ghosh & Tewari (IIT Roorkee)

What I reproduced

- Figs. 6 to 13: property coverage, the ZT and other property distributions, the compound type/crystal structure/doping breakdowns, and the ZT vs temperature trends for alloys vs oxides.
- Followed the method in their released `Post_processing_EDA.ipynb`, run on their publicly available dataset.

Why the numbers differ slightly

The paper's Figs. 6 to 13 are computed on their full dataset of 18,869 records. What they actually release publicly is a smaller subset of 11,927 entries that's been chemically validated (using `pymatgen`, to drop non-standard material names like “bismuth telluride” or “PbTe thin film”). That validated subset is what I had access to, so my numbers follow the same trends and the same method, just not identical to theirs.

Metric	Paper (18,869)	Reproduced (11,927)
Electrical resistivity coverage	11.8%	11.7%
Mean ZT	0.742	0.824*
Median ZT	0.720	0.750

**After removing a small number of non-physical ZT values (around 400 to 1000, well outside any plausible range) that were sitting in the data and had pulled the raw mean up to 1.355 before I filtered them out.*

Paper 2: Athar, Mecibah & Jund (Montpellier/ICGM)

What I reproduced

I used the authors' actual GitLab script and their actual `Starrydata2` export to run the round-robin bin filtering method on the three worked examples they use in the paper itself: `TiNiSn`, `NbFeSb`, and `TiCoSb`.

System	Paper	Reproduced
<code>TiNiSn</code> , selected bin (unique DOIs)	11	11
<code>TiNiSn</code> , reference <code>zT_max</code>	0.376	0.376
<code>NbFeSb</code> , mislabeled curves excluded	Yes	Yes
<code>TiCoSb</code> , mislabeled curves excluded	Yes	Yes

`TiNiSn` matches the paper exactly, including the same reference DOI and the same reference curve. `NbFeSb` and `TiCoSb` are the paper's own examples of wrongly attributed `Starrydata2` curves, and in both cases the method correctly picked out the genuine data and excluded the mislabeled curves, the same way the paper describes it.

What I couldn't reproduce, and why

The full dataset scale numbers (108 materials, 256 compositions, 1521 data points for ICGM, against Starrydata2's 132, 367, and 9003) need a file called `candidates.csv`, which the script reads to automatically work out which of the roughly 19,000 Starrydata2 curves are genuine half-Heuslers. I looked for this file in the GitLab repository, the paper's supplementary material, and its data availability statement, and it is not in any of them. I kept the reproduction to the three systems the paper itself uses as worked examples.